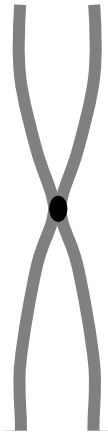
SCIENCE 1.3 **Name:** ***ANSWERS***

WORKSHEET FOUR **GENETIC MATERIAL**

1. Genes are called units of ***inheritance***
2. Genes are short sections of a very long molecule called ***DNA***
3. This long molecule has a shape called a double ***helix***
4. The complete DNA structure is called a C***hromosome***
5. In the nucleus of the human cheek cell there are ***46*** chromosomes.
6. Instead of giving the total number of chromosomes, we often say there are ***23*** pairs of chromosomes in the nucleus of a human cell.
7. Every gene codes for a particular ***trait, or characteristic (protein?)***
8. Because the chromosomes come in pairs, there are ***2*** genes coding for the same trait.
9. These two different forms of the same gene are called ***alleles***
10. Human eggs and sperm contain ***23*** chromosomes.
11. Genes consist of a sequence of ***bases*** on the DNA molecule.
12. Chromosomes become visible in the nucleus when the cell is ready to ***divide***
13. Match up the terms with their appropriate descriptors.

(a) DNA	**4**	(1) One half of a replicated chromosome.
(b) Gene	**3**	(2) The state of a nucleus when it has two sets of chromosomes.
(c) Diploid	**2**	(3) The unit of inheritance.
(d) Chromosome	**5**	(4) A double helix molecule that contains genes.
(e) Chromatid	**1**	(5) Long strands of DNA found in the cell nucleus.

14. 

(a) The replicated chromosome shown, consists of ***2*** chromatids joined together at the ***centromere***. These two strands of DNA are called "sister" chromatids because they contain identical sets of ***genes***

(b) DNA is the only molecule that can ***replicate*** itself.

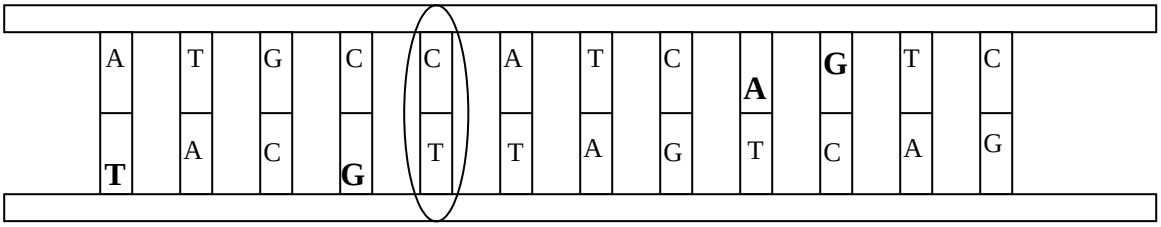
(c) To do this, it "unzips" and new nucleotides are ***added/matched*** to each side.

15. The two sides of a DNA strand are joined by pairs of ***bases***.

16. The bases are adenine, thymine, cytosine and guanine. These four bases are represented by letters ***A,T,C,G***.

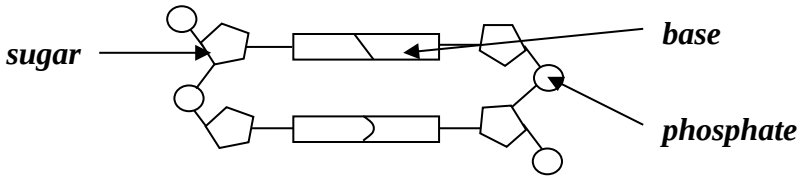
17. In the pairing up process adenine can only bond onto ***thymine*** and cytosine can only bond onto ***guanine***.

18. Complete the base sequence by labelling the unmarked bases with their correct letters.



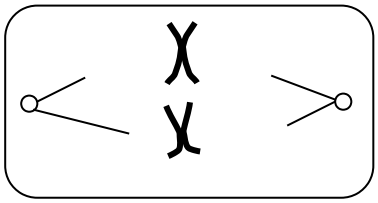
Circle the pair of bases which represents a **mutation.**

19. The structure shown is part of a DNA molecule. Draw three labels showing which part of it is a sugar, which part is a phosphate and which part of it is a base.



20. Cell division that produces identical cells is called ***mitosis***

21. In the process of mitosis, one of the steps can be drawn as shown. Describe what is happening at this stage of the cell division. ***The replicated chromosomes are lined up on the equator and the spindle fibres are growing towards them.***



22. Normal cell division produces new cells which are an exact copy of the parent cell. Suggest two reasons why this type of cell division needs to occur in a living organism.

For growth and repair

23. What type of cell division will replace the skin cells damaged when your knee is grazed in a fall? ***Mitosis***

24. A muscle cell in a human has ***46*** chromosomes. The cell replicates by mitosis forming two new daughter cells. The daughter cells will have ***46*** chromosomes.